

Example 1 : To solve the quadratic equation $ax^2 + bx + c = 0$, where $a \neq 0$, we can use the quadratic formula:

If $ax^2 + bx + c = 0$ and $a \neq 0$, then $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

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Proof:

1. Start with the quadratic equation $ax^2 + bx + c = 0$, where $a \neq 0$.

2. Divide the entire equation by a to simplify the equation:

$$x^2 + \left(\frac{b}{a}\right)x + \frac{c}{a} = 0$$

3. Move the constant term (c/a) to the right side of the equation:

$$x^2 + \left(\frac{b}{a}\right)x = -\frac{c}{a}$$

4. Add $(b/2a)^2$ to both sides of the equation to complete the square:

$$x^2 + \left(\frac{b}{a}\right)x + \left(\frac{b}{2a}\right)^2 = \left(\frac{b}{2a}\right)^2 - \frac{c}{a}$$

5. Simplify the right side of the equation:

$$x^2 + \left(\frac{b}{a}\right)x + \left(\frac{b^2}{4a^2}\right) = \frac{b^2 - 4ac}{4a^2}$$

6. Rewrite the left side of the equation as a perfect square:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

7. Take the square root of both sides of the equation:

$$x + \frac{b}{2a} = \pm \sqrt{\frac{b^2 - 4ac}{4a^2}}$$

8. Simplify the square root:

$$x + \frac{b}{2a} = \pm \frac{\sqrt{b^2 - 4ac}}{2a}$$

9. Subtract $b/2a$ from both sides of the equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

10. This is the quadratic formula, which provides the solutions for x in terms of a , b , and c .

Therefore, we have proven that if $ax^2 + bx + c = 0$ and $a \neq 0$, then the solution for x is given by $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ using the quadratic formula.

If n is an even integer greater than 2, then n is a sum of two primes.

Assumption: Let's assume that P is true, where P represents the statement " n is an even integer greater than 2."

To prove: We need to show that Q logically follows from the assumption, where Q represents the statement " n is a sum of two primes."

Proof:

1. Assume P : Let n be an even integer greater than 2.
2. Divide n by 2: Since n is an even integer, it can be written as $n = 2k$, where k is an integer.
3. Express n as the sum of two primes: Consider the cases where k is an even or odd integer.

Case 1: k is an even integer:

In this case, k can be written as $k = 2m$, where m is an integer.

Substituting this into $n = 2k$, we get $n = 2(2m) = 4m$.

Since $4m$ is divisible by 2, it is not a prime number.

However, $4m$ can be expressed as the sum of two primes, namely $2m + 2m$, where both $2m$ terms are prime.

Therefore, $n = 4m$ is a sum of two primes.

Case 2: k is an odd integer:

In this case, k can be written as $k = 2m + 1$, where m is an integer.

Substituting this into $n = 2k$, we get $n = 2(2m + 1) = 4m + 2$.

Since $4m + 2$ is divisible by 2, it is not a prime number.

However, $4m + 2$ can be expressed as the sum of two primes, namely $(2m + 1) + (2m + 1)$, where both $(2m + 1)$ terms are prime.

Therefore, $n = 4m + 2$ is a sum of two primes.

4. Conclusion: In both cases, we have shown that n can be expressed as the sum of two primes. Therefore, if n is an even integer greater than 2 (P), then n is a sum of two primes (Q).

By assuming P and logically deducing Q , we have proven the statement "If n is an even integer greater than 2, then n is a sum of two primes" using the method of assumption and logical deduction

Example: For all integers a and b , if a and b are positive and a divides b , then $a \leq b$

1. Assume a and b are positive integers and a divides b .
2. By definition, if a divides b , there exists an integer k such that $b = ak$.
3. Since a and b are positive integers, k must also be positive.
4. Multiply both sides of the equation $b = ak$ by a to obtain $ab = a^2k$.
5. Since a and k are both positive, a^2k is positive.
6. Dividing both sides of the equation $ab = a^2k$ by a (which is positive) gives $b = ak$.

7. Since $a \leq ak$ for any positive integer k , it follows that $a \leq b$.
8. Conclusion: If a and b are positive integers and a divides b , then $a \leq b$.

Example: For all integers a , b , and c , if a divides b and b divides c , then a divides c

1. Assume a , b , and c are integers such that a divides b and b divides c .
2. By definition, if a divides b , there exists an integer k such that $b = ak$.
3. Similarly, if b divides c , there exists an integer m such that $c = bm$.
4. Substitute the value of b from step 2 into the equation in step 3: $c = (ak)m$.
5. Simplify the equation: $c = a(km)$.
6. Since k and m are integers, km is also an integer.
7. Therefore, we can rewrite the equation as $c = a * \text{integer}$, showing that a divides c .
8. Conclusion: If a divides b and b divides c , then a divides c for all integers a , b , and c .

Example: Any two consecutive integers have opposite parity.

1. Assume that n and $n+1$ are two consecutive integers.
2. By definition, parity refers to whether a number is even or odd.
3. Case 1: Assume n is even.
4. By the definition of an even number, n can be expressed as $n = 2k$, where k is an integer.
5. Case 2: Assume n is odd.
6. By the definition of an odd number, n can be expressed as $n = 2k+1$, where k is an integer.
7. Adding 1 to both sides of the equation in case 6, we get $n+1 = 2k+1+1 = 2k+2 = 2(k+1)$.
8. In case 3, we expressed n as $2k$, which means it is even.
9. In case 7, we expressed $n+1$ as $2(k+1)$, which means it is also even.
10. Conclusion: In both cases, one of the consecutive integers is even and the other is odd, proving that any two consecutive integers have opposite parity.

To prove that $\sqrt{2}$ is irrational using the method of proof by contradiction, we can assume the opposite of the statement, which is that $\sqrt{2}$ is rational.

Assume that $\sqrt{2}$ is rational: This means that $\sqrt{2}$ can be expressed as the ratio of two integers, p and q , where q is not equal to zero, and p and q have no common factors other than 1.

If $\sqrt{2}$ is rational, then we can write it as $\sqrt{2} = p/q$.

Now, let's manipulate this equation to derive a contradiction.

Squaring both sides of the equation, we get $2 = p^2/q^2$.

This implies that $2q^2 = p^2$.

This means that p^2 is an even number since it is divisible by 2. Let's represent p^2 as $2k$, where k is an integer.

Substituting $2k$ for p^2 in the equation, we have $2q^2 = 2k$.

Dividing both sides of the equation by 2, we get $q^2 = k$.

This means that q^2 is also an even number since it is equal to k , which is an integer.

However, **this contradicts our initial assumption that p and q have no common factors other than 1. If both p^2 and q^2 are even**, they have a common factor of 2, which contradicts our assumption.

Since our assumption that $\sqrt{2}$ is rational leads to a contradiction, we can conclude that $\sqrt{2}$ is not rational, i.e., $\sqrt{2}$ is irrational.

Note: The proof by contradiction method shows that if assuming the opposite of a statement leads to a contradiction, then the original statement must be true.